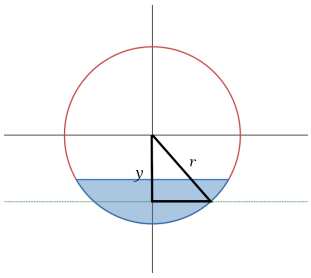
Problem 3. Suppose a bowl in the shape of a half-sphere contains water coming up to half of its height. What fraction of its *volume* is filled?

Solution. The volume of the bowl is half the volume of the sphere:

$$\frac{1}{2} \cdot \frac{4\pi r^3}{3} = \frac{2\pi r^3}{3}$$

We can compute the volume of water as an integral in y of the area $A(y)$ of a horizontal cross section. The following diagram gives a side-on view:



Using the Pythagorean Theorem, we see that the cross section of the water at height y is a disk of radius $\sqrt{r^2 - y^2}$, and so

$$A(y) = \pi(r^2 - y^2)$$

The vertical extent of the water is from $y = -r$ to $y = -r/2$, and so the volume is

$$V = \int_{-r}^{-r/2} A(y)\mathrm{d}y = \int_{-r}^{-r/2} \pi(r^2 - y^2)\mathrm{d}y$$

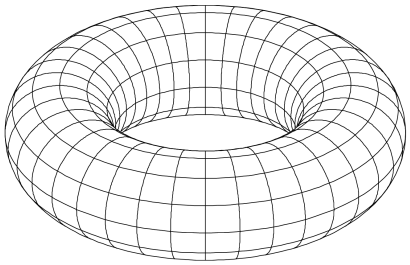
Using the substitution $u = -y$, and swapping the bounds of integration, this is

$$\pi \int_{r/2}^{r} \left[r^2 - u^2\right] \mathrm{d}u = \pi \left[r^2 u - \frac{u^3}{3}\right]\Bigg|_{r/2}^{r} = \pi\left(r^3 - \frac{r^3}{3} - \frac{r^3}{2} + \frac{r^3}{24}\right) = \frac{5\pi r^3}{24}$$

Comparing this with the volume of the bowl, the fraction of the volume which is filled is

$$\frac{\frac{5\pi r^3}{24}}{\frac{2\pi r^3}{3}} = \frac{5}{16}$$

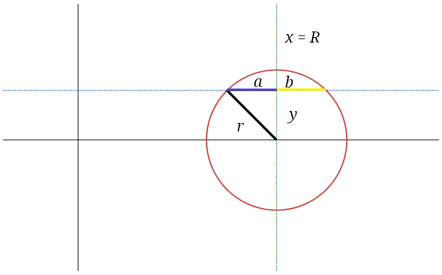
Problem 4 (Optional)**.** A torus is a solid obtained by revolving a disk around an axis. Letting $0 < r < R$



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be two radii, find the volume of the torus obtained by revolving the disk of radius r centered at $(R, 0)$ in the x, y-plane about the y-axis.

Solution. We will compute the volume as in integral in y of the area $A(y)$ of a horizontal cross section of height y. The cross section is a washer whose inner and outer radii come from the left and right sides of the disk being revolved. In the above diagram, the inner and outer radii are respectively $R - a$ and $R + b$. Using



the Pythagorean Theorem, we have $a = b = \sqrt{r^2 - y^2}$, and so our inner and outer radii are $R - \sqrt{r^2 - y^2}$ and $R + \sqrt{r^2 - y^2}$. The cross sectional area is then computed as follows:

$$A(y) = \pi((R + \sqrt{r^2 - y^2})^2 - (R - \sqrt{r^2 - y^2})^2)$$

Using the difference of squares factorization, this is

$$\pi(2R \cdot 2\sqrt{r^2 - y^2}).$$

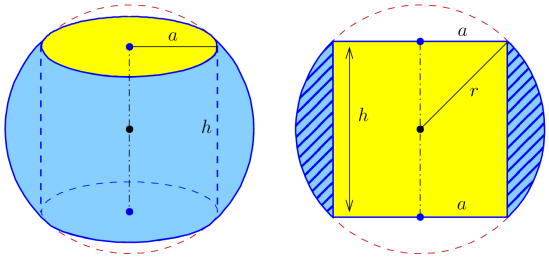
The vertical extent of the torus is from $y = -r$ to $y = r$, so the volume is

$$V = \int_{-r}^{r} A(y)\mathrm{d}y = \int_{-r}^{r} 4\pi R\sqrt{r^2 - y^2}\mathrm{d}y = 4\pi R \int_{-r}^{r} \sqrt{r^2 - y^2}\mathrm{d}y$$

The integral here is the area below the graph of $f(y) = \sqrt{r^2 - y^2}$, which is a half disk of radius r, and so has area $\pi r^2/2$. Thus, the volume comes out to

$$4\pi R(\pi r^2/2) = 2\pi R \cdot \pi r^2$$

Problem 5. Consider a ring obtained by drilling a cylinder out of a sphere of radius r, so that the height of the resulting solid is $h < 2r$.



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(a) Use the Pythagorean Theorem to find the radius a of the center hole in terms of r and h.

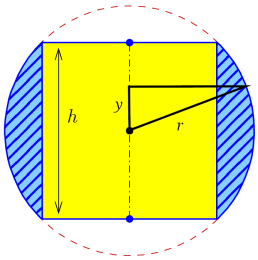
(b) Compute the volume of the ring. If you did this right, it should depend only on h and not on r. Isn't that strange?

Solution. (a) In the diagram above, we can see that a is one leg of right triangle whose other leg is $h/2$ and whose hypotenuse is r. The Pythagorean Theorem tells us that

$$a^2 + h^2/4 = r^2$$

and so $a = \sqrt{r^2 - h^2/4}$.

(b) We will compute the volume as in integral in y of $A(y)$, the area of a horizontal cross section at height y. The horizontal cross sections are washers whose inner radius is a and whose outer radius comes from the surface of the sphere. The right triangle depicted below tells us that the outer radius $R(y)$ satisfies $y^2 + R(y)^2 = r^2$, and so $R(y) = \sqrt{r^2 - y^2}$.



We can then compute the area of the cross section:

$$A(y) = \pi(R(y)^2 - a^2) = \pi(r^2 - y^2 - (r^2 - h^2/4)) = \pi(h^2/4 - y^2)$$

The vertical extend of the solid is from $y = -h/2$ to $y = h/2$ so the volume is computed as follows:

$$V = \int_{-h/2}^{h/2} A(y)\mathrm{d}y = \int_{-h/2}^{h/2} \pi(h^2/4 - y^2)\mathrm{d}y$$

Using symmetry and taking out the constant, this is

$$2\pi \int_0^{h/2} [h^2/4 - y^2]\mathrm{d}y = 2\pi \left[h^2 y/4 - y^3/3\right]\Big|_0^{h/2} = 2\pi \left[\frac{h^3}{8} - \frac{h^3}{24}\right] = 2\pi \frac{h^3}{12} = \frac{\pi h^3}{6}$$